Here's a **comprehensive theory summary** for **Session 6: Data Center Planning & Design**, integrating key aspects and industry best practices.

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1. Data Center Architecture, Requirements & Prerequisites

- The design must support **scalability**, **power**, **cooling**, **networking**, **physical security**, and **resiliency**, aligned with the data center's intended usage (e.g., cloud, HPC) (tierpoint.com, cc-techgroup.com).
- Standards like **Uptime Institute Tiers**, **TIA-942**, **ANSI/BICSI 002**, **ISO/IEC 22237**, and **EN 50600** define requirements ranging from site selection to redundancy and sustainability (gbc-engineers.com).

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2. Required Physical Area & Unoccupied Space

- Estimate space per rack, including hot/cold aisles and infrastructure.
- Example: ~20 sq ft per rack (12 sq ft footprint + access space) .

- Leave room for future expansion (~70–80% hall utilization; 20–30% support areas) (reddit.com).

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3. Power Requirements

- Determine power load per rack—typically 3–4 kW potentially up to 15 kW in dense setups (reddit.com).
- Provide dual feeds via UPS and generators with redundancy matching tier level.
- **Power Usage Effectiveness (PUE)** measures energy efficiency; best-in-class systems achieve PUE ~1.1–1.2 (tierpoint.com, en.wikipedia.org).

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4. Cooling & HVAC

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Use CRAC/CRAH units, hot-aisle/cold-aisle containment, and possibly immersion or in-row cooling (csemag.com).

- Design for ~2 kW per rack (gross area), scaling to 10 kW in high-density zones (docs.oracle.com).

- Maintain environmental conditions within ASHRAE recommendations: 8–27 °C and 50–70 %RH (kslaw.com).

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5. Structural Considerations

- Raised floors provide cable routing and support underfloor cooling. Panels must support ~12 kN/m² uniformly and point loads ~4.5 kN (en.wikipedia.org).

- Weight capacity, ceiling height (~8 ft 6 in), and clear aisle space are critical planning factors (docs.oracle.com).

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6. Network Bandwidth & Connectivity

- Provide redundant high-speed connectivity (1–10 Gb+), carrier-neutral access, and backup routes for availability (netconglobal.com).
- Monitor bandwidth use (e.g. 95th percentile usage), with headroom for growth (reddit.com).

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7. Budget Constraints

- Data center TCO spans **construction (50–80%)**, **hardware (20–40%)**, and **operational (power, cooling, staff)**—requiring precise initial and lifecycle budgeting (netconglobal.com).

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8. Geographic Location Selection

- Choose low-risk areas (flood, earthquake, pollution), with reliable utilities and cooling climate (netconglobal.com).
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Requirements include inexpensive power/water, skilled local staff, low hazard exposure, and site suitability (e.g., no overhead windows, strong containment) .

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9. Existing Building Considerations

- Evaluate retrofits: structural load, floor height, HVAC capacity, utility access, airflow design, and security zones (cc-techgroup.com).

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10. Characteristics of Outstanding Design

- Emphasize modularity, scalability, redundancy, energy efficiency, sustainability (e.g., immersion cooling), and security by design .

11-12. Planning Guidelines & Structures

- Infrastructure must include MEP, fire suppression (dry/wet, gaseous options), environmental monitoring, security, and BMS integration .
- Common building structures: raised floors, containment modules/containers, modular blocks, secure zones (perimeter fencing, mantraps) (en.wikipedia.org).

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13. Raised Floor Design & Deployment

- Ensure correct pedestal height for airflow and cabling.
- Select panels suited to load, seal vapour barriers, align airflow design with cold aisle/hot aisle, and carry out ongoing inspections for structural integrity .

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14. Protection Against Vandalism

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Implement layered physical security: fencing, bollards, surveillance, controlled access, no-grade-level windows, protected external units, and perimeter planning (en.wikipedia.org).

Summary Table

Consideration	Key Guidelines
Space & Layout	20 sq ft/rack, aisles, raised floors, future expansion
Power	3–15 kW/rack, UPS/generator redundancy, aim PUE ~1.1–1.2
Cooling	CRAC/CRAH, containment, immersion/in-row options, ASHRAE conditions
Structure	Raised floors with proper load rating, ceiling height, room layout
Network	Redundant, carrier-neutral, high-bandwidth links
Site Selection	Low-risk, affordable utilities, talent availability
Security	Layered zones, surveillance, bollards, locked racks
Standards	Adhere to TIA-942, Uptime, BICSI, ISO, EN for reliability & sustainability

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Once you're ready, I can create a set of 40 MCQs (10 easy, 20 medium, 10 hard) aligned with this theory.